# Class 9 Mathematics — Exercise 3.4

## Basic Concept

### (1) Binary Relation

A binary relation $R$ from set $A$ to set $B$ is **any subset** of the Cartesian product $A \times B$, i.e. $R \subseteq A \times B$.

- If $(x, y) \in R$, we write $x\,R\,y$ ($x$ is related to $y$).
- A relation in $A$ (from $A$ to $A$) is a subset of $A \times A$.

### (2) Number of Binary Relations

If $n(A \times B) = m$, then the number of binary relations from $A$ to $B = 2^m = 2^{n(A)\times n(B)}$.

### (3) Domain and Range

- **Domain** of $R$ = set of all **first** elements of the ordered pairs of $R$.
- **Range** of $R$ = set of all **second** elements of the ordered pairs of $R$.

### (4) Inverse Relation

$R^{-1}$ is obtained by **interchanging** the elements of every ordered pair of $R$:

$$R^{-1} = \{(y, x) \mid (x, y) \in R\}, \qquad \text{Dom } R^{-1} = \text{Range } R, \qquad \text{Range } R^{-1} = \text{Dom } R$$

## Question 1

**Find the number of binary relations in the following cases.**

**(i)** $A = \{1, 3\}$, $B = \{0, 2, 4\}$

$n(A \times B) = n(A) \times n(B) = 2 \times 3 = 6$

$$\text{Number of binary relations} = 2^6 = \boxed{64}$$

**(ii)** $n(C) = 7$ (relations in $C$)

$n(C \times C) = 7 \times 7 = 49$

$$\text{Number of binary relations} = \boxed{2^{49}}$$

**(iii)** $D = \{1, 3, 5\}$ (relations in $D$)

$n(D \times D) = 3 \times 3 = 9$

$$\text{Number of binary relations} = 2^9 = \boxed{512}$$

## Question 2

**Find all possible binary relations in the following cases mentioning the number of binary relations in each case.**

**(i)** $A = \{\sqrt{2}, \sqrt{3}, \sqrt{5}\}$, $B = \{\sqrt[3]{5}\}$ (relations from $A$ to $B$)

$$\text{Cartesian product} = \{(\sqrt{2}, \sqrt[3]{5}), (\sqrt{3}, \sqrt[3]{5}), (\sqrt{5}, \sqrt[3]{5})\}$$

Number of elements $= 3$, number of binary relations $= 2^3 = \boxed{8}$

$R_1 = \{\ \}$
$R_2 = \{(\sqrt{2}, \sqrt[3]{5})\}$
$R_3 = \{(\sqrt{3}, \sqrt[3]{5})\}$
$R_4 = \{(\sqrt{5}, \sqrt[3]{5})\}$
$R_5 = \{(\sqrt{2}, \sqrt[3]{5}), (\sqrt{3}, \sqrt[3]{5})\}$
$R_6 = \{(\sqrt{2}, \sqrt[3]{5}), (\sqrt{5}, \sqrt[3]{5})\}$
$R_7 = \{(\sqrt{3}, \sqrt[3]{5}), (\sqrt{5}, \sqrt[3]{5})\}$
$R_8 = \{(\sqrt{2}, \sqrt[3]{5}), (\sqrt{3}, \sqrt[3]{5}), (\sqrt{5}, \sqrt[3]{5})\}$

**(ii)** $C = \{\pi, e\}$ (relations in $C$)

$$\text{Cartesian product} = \{(\pi, \pi), (\pi, e), (e, \pi), (e, e)\}$$

Number of elements $= 4$, number of binary relations $= 2^4 = \boxed{16}$

$R_1 = \{\ \}$
$R_2 = \{(\pi, \pi)\}$
$R_3 = \{(\pi, e)\}$
$R_4 = \{(e, \pi)\}$
$R_5 = \{(e, e)\}$
$R_6 = \{(\pi, \pi), (\pi, e)\}$
$R_7 = \{(\pi, \pi), (e, \pi)\}$
$R_8 = \{(\pi, \pi), (e, e)\}$
$R_9 = \{(\pi, e), (e, \pi)\}$
$R_{10} = \{(\pi, e), (e, e)\}$
$R_{11} = \{(e, \pi), (e, e)\}$
$R_{12} = \{(\pi, \pi), (\pi, e), (e, \pi)\}$
$R_{13} = \{(\pi, \pi), (\pi, e), (e, e)\}$
$R_{14} = \{(\pi, \pi), (e, \pi), (e, e)\}$
$R_{15} = \{(\pi, e), (e, \pi), (e, e)\}$
$R_{16} = \{(\pi, \pi), (\pi, e), (e, \pi), (e, e)\}$

**(iii)** $D = \{5\}$, $E = \{1, 10\}$ (relations from $D$ to $E$)

$$\text{Cartesian product} = \{(5, 1), (5, 10)\}$$

Number of elements $= 2$, number of binary relations $= 2^2 = \boxed{4}$

$R_1 = \{\ \}$
$R_2 = \{(5, 1)\}$
$R_3 = \{(5, 10)\}$
$R_4 = \{(5, 1), (5, 10)\}$

## Question 3

**If $P = \{7, 8, 9\}$ then find 2 binary relations from $P$ to $P$. Also find domain and range of each relation.**

$$P \times P = \{(7,7), (7,8), (7,9), (8,7), (8,8), (8,9), (9,7), (9,8), (9,9)\}$$

Any subset of $P \times P$ is a relation. For example:

$$R_1 = \{(7,7), (7,8)\}, \qquad \text{Dom } R_1 = \{7\}, \qquad \text{Range } R_1 = \{7,8\}$$

$$R_2 = \{(7,9), (8,7), (8,9)\}, \qquad \text{Dom } R_2 = \{7,8\}, \qquad \text{Range } R_2 = \{7,9\}$$

## Question 4

**Let $H = \{5, 6, 7, 8, 9\}$ and $G = \{5, 7, 9, 11\}$. Write the following relations from $H$ to $G$ in tabular form.**

**(i)** 'is equal to': $R = \{(5,5), (7,7), (9,9)\}$

**(ii)** 'is less than': $R = \{(5,7), (5,9), (5,11), (6,7), (6,9), (6,11), (7,9), (7,11), (8,9), (8,11), (9,11)\}$

**(iii)** 'is greater than': $R = \{(6,5), (7,5), (8,5), (8,7), (9,5), (9,7)\}$

**(iv)** 'is one less than': $R = \{(6,7), (8,9)\}$

**(v)** 'is one greater than': $R = \{(6,5), (8,7)\}$

**(vi)** 'is two less than': $R = \{(5,7), (7,9), (9,11)\}$

## Question 5

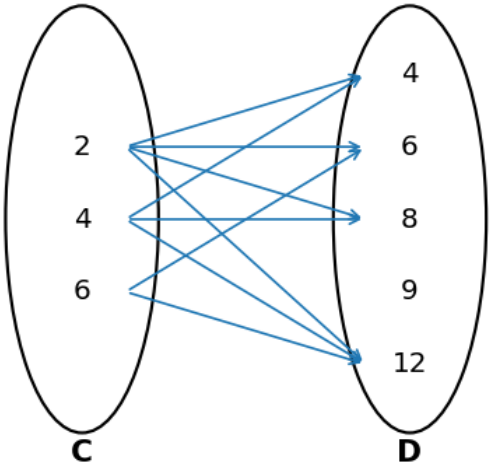
**Let $C = \{2, 4, 6\}$, $D = \{4, 6, 8, 9, 12\}$ and $R = \{(x, y) \mid x \in C, y \in D \wedge x$ is factor of $y\}$. (i) Write $R$ in tabular form. (ii) Find domain and range of $R$. (iii) Find $R^{-1}$. (iv) Represent $R$ by arrow diagram.**

**(i)** $R = \{(2,4), (2,6), (2,8), (2,12), (4,4), (4,8), (4,12), (6,6), (6,12)\}$

**(ii)** Dom $R = \{2, 4, 6\}$, Range $R = \{4, 6, 8, 12\}$

**(iii)** $R^{-1} = \{(4,2), (4,4), (6,2), (6,6), (8,2), (8,4), (12,2), (12,4), (12,6)\}$

**(iv)** Arrow diagram of $R$:



## Question 6

**Let $R = \{(2,0),(4,2),(6,4),(8,6),(10,8)\}$. (i) Write $R$ in set builder form. (ii) Find domain and range of $R$. (iii) Write $R^{-1}$ in tabular and set builder form. (iv) Represent $R$ and $R^{-1}$ by arrow diagram.**

In every pair, the first element is 2 more than the second element: $x - y = 2$.

**(i)** $R = \{(x,y) \mid x - y = 2\}$ (where $x \in \{2,4,6,8,10\}$)

**(ii)** Dom $R = \{2,4,6,8,10\}$, Range $R = \{0,2,4,6,8\}$

**(iii)** Tabular form: $R^{-1} = \{(0,2),(2,4),(4,6),(6,8),(8,10)\}$

Set builder form: $R^{-1} = \{(x,y) \mid y - x = 2\}$

**(iv)** Arrow diagrams of $R$ and $R^{-1}$:




## Question 7

**Let $A = \{0,1,3\}$ and $B = \{1,2,3,5,7\}$. Write $R = \{(x,y) \mid x \in A, y \in B \wedge y = 2x+1\}$ in tabular form. Also find $R^{-1}$.**

Put $x = 0$: $y = 2(0)+1 = 1 \in B$ ✓; $x = 1$: $y = 3 \in B$ ✓; $x = 3$: $y = 7 \in B$ ✓

$$R = \{(0,1),(1,3),(3,7)\}, \qquad R^{-1} = \{(1,0),(3,1),(7,3)\}$$

## Questions 8 and 9

**If $S = \{1,2,4,8\}$, $T = \{3^0, 3^1, 3^2\}$, then write the following binary relations in tabular form and find the domain and range in each case.**

Note: $3^0 = 1$, $3^1 = 3$, $3^2 = 9$, so $T = \{1,3,9\}$.

**(i)** $R_1 = \{(x,y) \mid x \in S, y \in T \wedge x = y\}$

$$R_1 = \{(1,3^0)\}$$

Dom $R_1 = \{1\}$, Range $R_1 = \{3^0\}$

**(ii)** $R_2 = \{(x,y) \mid x \in S, y \in T \wedge y < x\}$

$$R_2 = \{(2,3^0),(4,3^0),(4,3^1),(8,3^0),(8,3^1)\}$$

Dom $R_2 = \{2,4,8\}$, Range $R_2 = \{3^0, 3^1\}$

**(iii)** $R_3 = \{(x, y) \mid x \in S, y \in T \wedge x + y \in E\}$

$$R_3 = \{(1, 3^0), (1, 3^1), (1, 3^2)\}$$

Dom $R_3 = \{1\}$, Range $R_3 = \{3^0, 3^1, 3^2\}$

**(iv)** $R_4 = \{(x, y) \mid x \in S, y \in T \wedge x \times y \in O\}$

$$R_4 = \{(1, 3^0), (1, 3^1), (1, 3^2)\}$$

Dom $R_4 = \{1\}$, Range $R_4 = \{3^0, 3^1, 3^2\}$

**(v)** $R_5 = \{(x, y) \mid x \in S, y \in T \wedge y > 2x\}$

$$R_5 = \{(1, 3^1), (1, 3^2), (2, 3^2), (4, 3^2)\}$$

Dom $R_5 = \{1, 2, 4\}$, Range $R_5 = \{3^1, 3^2\}$